

EXECUTIVE SUMMARY

Executive Summary

Procure-to-Pay · Acme Manufacturing

REPORT REFERENCE	Executive Summary
CLIENT	Acme Manufacturing
DOMAIN	Procure-to-Pay
PREPARED BY	AuroPro · Autonomous Discovery Platform
CLASSIFICATION	Confidential

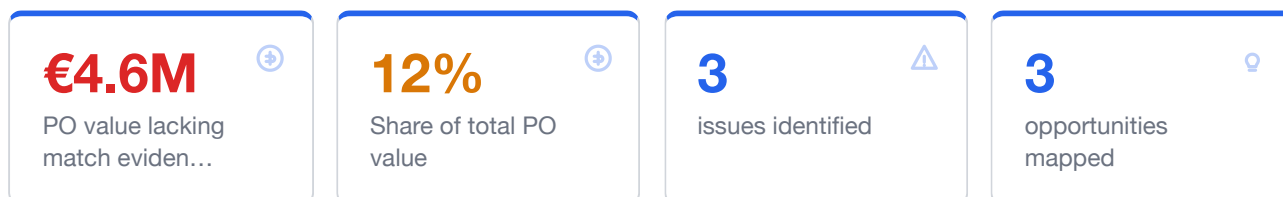
Contents

1	At a glance
1.1	Situation & opportunity
2	Where to start
3	What we read

Executive Summary

Seven purchase orders worth EUR 557,000 — 12% of total PO value — were released to suppliers on a single approval, in breach of the dual-approval rule that applies above the EUR 50,000 threshold.

1 At a glance



1.1 Situation & opportunity

The situation

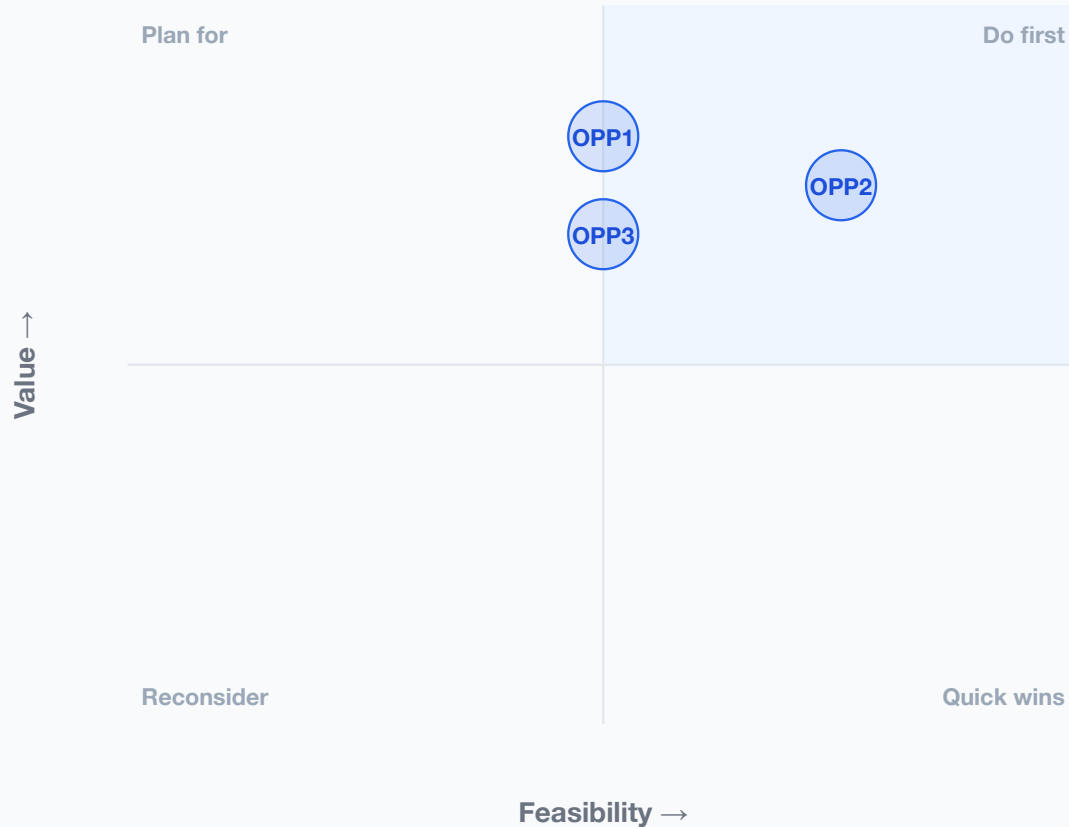
A review of the purchase-order export against the procurement policy surfaced three distinct control gaps. First, 7 POs totalling EUR 557,000 (12% of total PO value) cleared on single approval only, despite every one of these orders exceeding the EUR 50,000 limit that requires both Category Manager and Finance sign-off before release. Second, 3 POs worth EUR 216,000 (4.7% of total PO value) were placed as maverick spend — goods ordered without a prior purchase order — which the policy prohibits under any circumstances; 2 of these were raised by a single buyer (buyer_a). Third, the export carries no goods-receipt or invoice fields, leaving EUR 4,625,800 of PO value without evidence of the required three-way match.

The opportunity

The concentration of exposure is clear and addressable: EUR 557,000 of single-approval orders and EUR 216,000 of maverick spend together represent the most immediate compliance risk, with maverick spend further concentrated in one buyer. The priority is to remediate the dual-approval breaches on the high-value orders, address the maverick-spend pattern at the buyer level, and close the visibility gap by capturing goods-receipt and invoice data so the EUR 4,625,800 in unmatched PO value can be three-way matched.

2 Where to start

WHERE EACH OPPORTUNITY SITS ON VALUE VERSUS FEASIBILITY



HITL WORKFLOW

Address: Three-way match evidence cannot be verified across the PO population

The procurement policy requires a three-way match — purchase order, goods receipt, and supplier invoice — before an invoice is approved for payment. ...

HITL WORKFLOW

Address: High-value purchase orders released on a single approval

High-value purchase orders are being released into the order pipeline on the strength of a single approval, bypassing the segregation-of-duties contr...

3 What we read

Purchase Order Export, Procurement Policy. Every figure in this assessment is computed from these sources and traces back to them.

Read on: the Current State, the issues found, and the recommended opportunities.

REPORT 01 OF 06

Current State Assessment

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Contents

1	Domain overview
1.1	Context and scope
1.2	Volume baseline
2	Process flow
3	Process inventory
4	Ownership map
4.1	Process inventory summary
4.2	RACI matrix
5	System inventory
5.1	System profiles
5.2	Information format & structure
6	Handoff map

Current State Assessment

How Procure-to-Pay runs today at Acme Manufacturing. A factual baseline of the process, the systems that support it, and how information is structured — stated as fact, no judgements.

1 Domain overview

1.1 Context and scope

This section documents the current state of the procurement-to-purchase-order process as evidenced by the procurement policy and the purchase order export. The procurement policy sets out four control statements governing how purchase orders are raised, approved, matched, and how ordering without a purchase order is treated. The purchase order export is the transactional record of issued purchase orders, carrying the fields `po_id`, `supplier`, `category`, `amount_eur`, `raised_by`, `approval_status`, and `po_before_order`.

Purchase orders are raised in the procurement system before goods are ordered. Orders above the fifty-thousand-euro threshold require approval from the Category Manager and a second approval from Finance before release to the supplier. Once goods are received and the supplier invoice arrives, the purchase order is matched against the goods receipt and the supplier invoice in a three-way match. The export covers buyers identified as `buyer_a` through `buyer_h` across categories including Components, Logistics, Raw Materials, Packaging, Services, and Consumables.

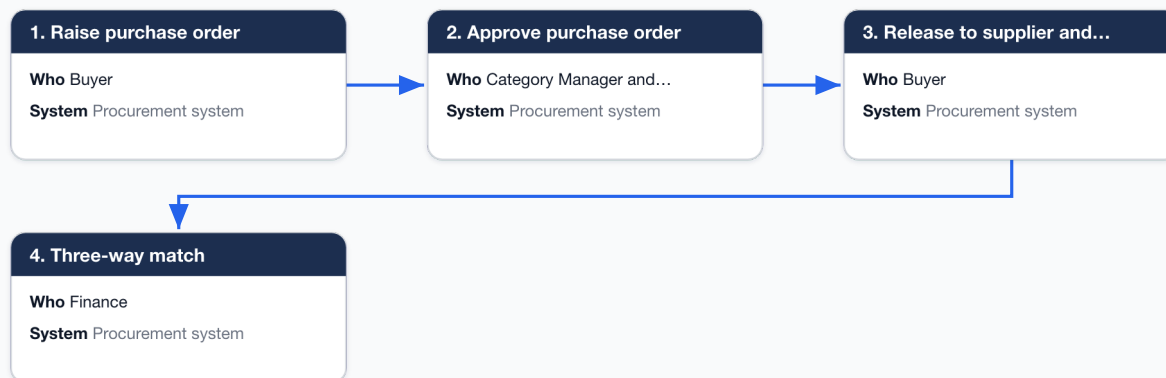
1.2 Volume baseline



2 Process flow

The end-to-end flow, who performs each step and on which system:

HOW WORK MOVES, END TO END



STEP	PERFORMED BY	SYSTEM	WHAT HAPPENS
1. Raise purchase order	Buyer	Procurement system	A buyer raises a purchase order in the procurement system. Policy states all purchase orders must be raised in the procurement system before goods are ordered. The export records three purchase orders where po_before_order is no, valued at EUR 216,000, with buyer_a accounting for two of them.
2. Approve purchase order	Category Manager and Finance	Procurement system	Purchase orders above the fifty-thousand-euro threshold require approval from the Category Manager and a second approval from Finance before release to the supplier. The export flags seven purchase orders as single_approval_only, valued at EUR 557,000, representing twelve percent of total purchase order value; all carry amounts above the threshold.
3. Release to supplier and order goods	Buyer	Procurement system	Following approval, the purchase order is released to the supplier and goods are ordered against it.
4. Three-way match	Finance	Procurement system	Purchase orders are matched against the goods receipt and the supplier invoice in a three-way match. The export carries the columns po_id, supplier, category, amount_eur, raised_by, approval_status, and po_before_order, with no goods_receipt or invoice column present. Total purchase order value lacking match evidence is EUR 4,625,800.

3 Process inventory

Each stage of the process in detail — how it runs today, who performs it, and on which system.

Raising purchase orders

Buyer · Procurement system

Policy statement one requires all purchase orders to be raised in the procurement system before goods are ordered. The export captures the `po_before_order` field, which records three purchase orders as no, summing to EUR 216,000; `buyer_a` accounts for two of these and `buyer_d` for one.

Source: Procurement Policy and Purchase Order Export

Approval thresholds

Category Manager; Finance · Procurement system

Policy statement two requires any purchase order above the fifty-thousand-euro threshold to receive approval from the Category Manager and a second approval from Finance before it is released to the supplier. The export flags seven purchase orders as `single_approval_only`, all with amounts above the threshold, totalling EUR 557,000 or twelve percent of total purchase order value.

Source: Procurement Policy and Purchase Order Export

Three-way matching

Finance · Procurement system

Policy statement three states purchase orders are matched against the goods receipt and the supplier invoice in a three-way match. The export columns are `po_id`, `supplier`, `category`, `amount_eur`, `raised_by`, `approval_status`, and `po_before_order`, with no `goods_receipt` or `invoice` column present. EUR 4,625,800 of purchase order value lacks match evidence.

Source: Procurement Policy and Purchase Order Export

Ordering without a purchase order

Buyer · Procurement system

Policy statement four states that maverick spend — ordering without a purchase order — is not permitted under any circumstances. The export records three such purchase orders valued at EUR 216,000, representing 4.7 percent of total purchase order value.

Source: Procurement Policy and Purchase Order Export

4 Ownership map

4.1 Process inventory summary

- **Purchase order raising** — Create a purchase order in the procurement system before goods are ordered.
- **Approval** — Apply Category Manager and Finance approval for orders above the threshold before release.
- **Three-way matching** — Match the purchase order against the goods receipt and the supplier invoice.

4.2 RACI matrix

ACTIVITY	RESPONSIBLE	ACCOUNTABLE
Raise purchase order in procurement system before ordering goods	Buyer	Buyer
First approval for POs above the threshold	Category Manager	Category Manager
Second approval for POs above the threshold before release to supplier	Finance	Finance
Three-way match against goods receipt and supplier invoice	Finance	Finance

5 System inventory

SYSTEM	ROLE	SYSTEM OF RECORD FOR
Procurement system	System in which all purchase orders must be raised before goods are ordered	Purchase orders, approvals, and three-way matching
Purchase order export	Transactional extract of issued purchase orders	po_id, supplier, category, amount_eur, raised_by, approval_status, po_before_order

5.1 System profiles

Each system that supports this process — what it is, how it is used, who owns it, and the constraints observed.

Procurement system

Records purchase orders, applies approval thresholds, and supports three-way matching as set out in the procurement policy.

How it's used. Buyers raise purchase orders before goods are ordered; Category Manager and Finance approve orders above the threshold; purchase orders are matched against goods receipt and supplier invoice.

Ownership & access. Buyers, Category Manager, Finance

Observed constraints. The export carries no goods_receipt or invoice column.

Purchase order export

Provides the transactional record of purchase orders for review.

How it's used. Holds the fields po_id, supplier, category, amount_eur, raised_by, approval_status, and po_before_order across buyers buyer_a through buyer_h.

Ownership & access. Procurement

Observed constraints. No goods_receipt or invoice column present.

5.2 Information format & structure

The patterns the source information follows — this drives how it can be ingested and reasoned over.

PATTERN	DESCRIPTION	WHERE IT APPEARS
approval_status = approved	Purchase orders recorded as approved in the export.	Suppliers include PackRight, CargoLink, PrecisionParts
single_approval_only	Purchase orders carrying a single approval, all with amounts above the threshold.	Seven POs totalling EUR 557,000
po_before_order = yes	Purchase order raised in the procurement system before goods were ordered.	Suppliers include ITPartner, FreightOne, ForgePrime
po_before_order = no	Goods ordered without a prior purchase order.	Three POs totalling EUR 216,000

6 Handoff map

How work crosses between steps and systems — each handoff is where information changes hands or format.



- **Raise purchase order → Approve purchase order** (*Procurement system routes POs above the threshold for Category Manager and Finance approval*)
- **Approve purchase order → Release to supplier and order goods** (*Release to supplier after second approval from Finance*)
- **Release to supplier and order goods → Three-way match** (*Purchase order matched against goods receipt and supplier invoice*)



REPORT 02 OF 06

Pain Points & Opportunities

Procure-to-Pay · Acme Manufacturing

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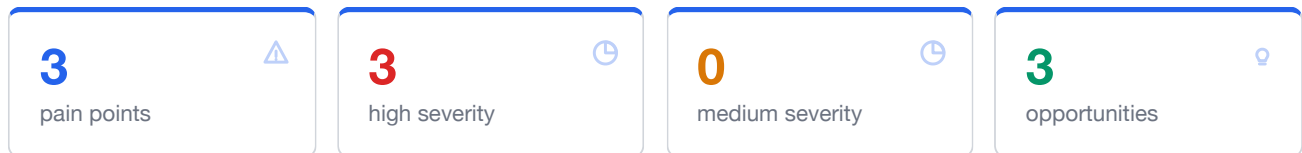
Contents

1	Issues at a glance
2	Root cause
3	Pain points in detail
4	Appendix — evidence register

Pain Points & Opportunities

The issues found in the discovery, ranked by business impact, each mapped to a recommended opportunity.

1 Issues at a glance



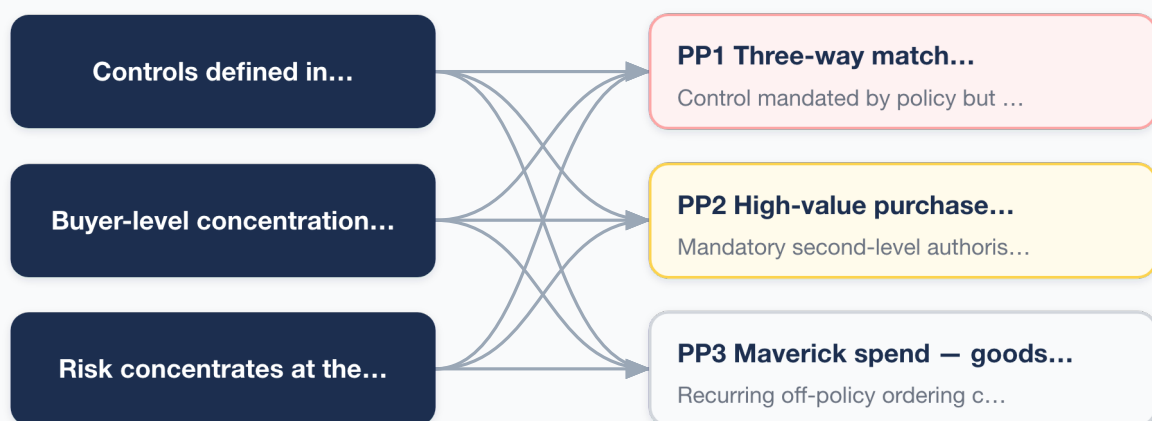
ISSUES RANKED BY BUSINESS IMPACT (MOST MATERIAL FIRST)



2 Root cause

How the issues found trace back to shared structural causes:

HOW THE ISSUES TRACE BACK TO SHARED CAUSES



Left: structural causes. Right: the pain points each one drives.

Controls defined in policy are not enforced or evidenced in the system. All three pain points share a common shape: the procurement policy specifies a control (three-way match, dual approval above EUR 50,000, no ordering without a PO), but the operational data either lacks the

fields to evidence it or shows it being bypassed. The gap is between written policy and system enforcement.

Buyer-level concentration of off-policy activity. Off-policy behaviour is not evenly spread. buyer_a alone accounts for two of the three maverick-spend cases, suggesting that targeted buyer coaching and controls could address a disproportionate share of the issue.

Risk concentrates at the high-value end. Single-approval breaches occur exclusively on orders above EUR 50,000, meaning the missing control coincides with the largest financial exposures rather than low-value transactions.

3 Pain points in detail

PP
01

Three-way match evidence cannot be verified across the PO population

HIGH SEVERITY

CONTROL EVIDENCE / FINANCIA...

Procurement policy requires every purchase order to be matched against the goods receipt and the supplier invoice (three-way match). However, the purchase order export carries no goods_receipt or invoice column, so there is no field-level evidence that any matching took place. As a result, a large body of PO value cannot be confirmed as matched.

€4.6M

PO VALUE LACKING MATCH EVIDENCE

Match evidence gap

POLICY REQUIREMENT	DATA FIELD EXPECTED	FIELD PRESENT?	VALUE AFFECTED (EUR)
Three-way match (PO vs goods receipt vs invoice)	goods_receipt	No	4,625,800
Three-way match (PO vs goods receipt vs invoice)	invoice	No	4,625,800

The export columns are po_id, supplier, category, amount_eur, raised_by, approval_status, po_before_order — no goods_receipt or invoice column present. Value affected is reported with Amber confidence.

Source: Procurement Policy and Purchase Order Export

Root cause: The system of record used to manage and export purchase orders does not capture goods receipt or invoice references, leaving the three-way match control unsupported by data.

BUSINESS IMPACT

Without match evidence, the organisation cannot confirm that goods were received or that invoices correspond to authorised orders, exposing it to overpayment, duplicate payment and undetected supplier billing errors.

ADDRESSED BY

OPP1 — see the Opportunity Portfolio.

Where this comes from: *Procurement Policy and Purchase Order Export*

PP
02

High-value purchase orders released on a single approval

HIGH SEVERITY

AUTHORISATION / SEGREGATION...

Policy requires any purchase order above EUR 50,000 to carry approval from the Category Manager AND a second approval from Finance before release to the supplier. Seven purchase orders, every one of them above the EUR 50,000 threshold, were released on a single approval only.

7

SINGLE-APPROVAL POS

€557K

VALUE OF SINGLE-APPROVAL POS

12%

SHARE OF TOTAL PO VALUE

Single-approval high-value orders against the EUR 50,000 threshold

METRIC	VALUE
POs flagged single_approval_only	7
Total value (EUR)	557,000
Share of total PO value	12%
Approval threshold breached (EUR)	above 50,000
Observed amounts (EUR)	54,000; 61,000; 67,000; 72,000; 88,000; 95,000; 120,000

All 7 single_approval_only rows have amounts above the EUR 50,000 dual-approval threshold.

Source: *Procurement Policy and Purchase Order Export*

Root cause: The dual-approval control for high-value orders is not enforced at the point of release, allowing orders above the threshold to proceed without the required second (Finance) sign-off.

BUSINESS IMPACT

High-value commitments are entered into without independent Finance challenge, increasing the risk of unbudgeted spend, pricing errors and fraud on the orders carrying the greatest financial exposure.

ADDRESSED BY

OPP3 — see the Opportunity Portfolio.

*Where this comes from: Procurement Policy and Purchase Order Export***PP
03****Maverick spend — goods ordered without a prior purchase order****HIGH SEVERITY****PROCESS COMPLIANCE / SPEND ...**

Policy states that all purchase orders must be raised in the procurement system before goods are ordered, and that maverick spend — ordering without a purchase order — is not permitted under any circumstances. Three purchase orders were recorded with po_before_order = no, indicating goods were ordered before the PO existed.

3

MAVERICK POS

€216K

MAVERICK SPEND VALUE

4.7%SHARE OF TOTAL PO
VALUE**2**MAVERICK POS BY
BUYER_A

Maverick spend by buyer

BUYER	MAVERICK PO COUNT	SUMMED VALUE (EUR)
buyer_a	2	
buyer_d	1	
Total	3	216,000

po_before_order=no: buyer_a count 2, buyer_d count 1; summed amount_eur 216,000. Per-buyer value split is not provided in the dataset.

Source: Procurement Policy and Purchase Order Export

Root cause: There is no preventive control stopping buyers from committing to suppliers before a PO is raised; the PO is created retrospectively after the order is placed.

BUSINESS IMPACT

Commitments are made outside the controlled procurement flow, removing the opportunity for budget checks, approval and negotiated pricing, and undermining supplier and spend governance.

ADDRESSED BY

OPP2 — see the Opportunity Portfolio.

Where this comes from: Procurement Policy and Purchase Order Export

4 Appendix — evidence register

Every finding traced to the source it rests on, with the confidence tier.

FINDING	SOURCE	EVIDENCE TYPE	KEY DATA POINT	CONFIDENCE
Three-way match is required by policy	procurement-policy	policy quote	"3. Purchase orders are matched against the goods receipt and the supplier invoice (three-way match)"	VERIFIED
Export has no fields to evidence matching	purchase-order-export	data structure	"Columns: po_id, supplier, category, amount_eur, raised_by, approval_status, po_before_order — no goods_receipt or invoice column present"	VERIFIED
PO value lacking match evidence	procurement-policy; purchase-order-export	computed measure	Total PO value lacking match evidence (EUR) = 4,625,800	AMBER
Orders above EUR 50,000 require dual approval	procurement-policy	policy quote	"2. Any purchase order above EUR 50,000 requires approval from the Category Manager AND a second approval from Finance before it is released to the supplier."	VERIFIED
Seven POs released on a single approval, all above threshold	purchase-order-export	data observation	"All 7 single_approval_only rows have amount_eur in {54000,61000,67000,72000,88000,95000,120000}, every value above the 50000 threshold"	VERIFIED
Single-approval value and share	procurement-policy; purchase-order-export	computed measure	Value EUR 557,000; 12% of total PO value	VERIFIED
Ordering requires a prior PO; maverick spend prohibited	procurement-policy	policy quote	"1. All purchase orders must be raised in the procurement system before goods are ordered." / "4. Maverick spend — ordering without a purchase order — is not permitted under any circumstances."	VERIFIED

FINDING	SOURCE	EVIDENCE TYPE	KEY DATA POINT	CONFIDENCE
Three maverick-spend POs by buyer, value EUR 216,000	purchase-order-export	data observation	"po_before_order=no: buyer_a count 2, buyer_d count 1; summed amount_eur 216000"	VERIFIED
Maverick spend share of total	procurement-policy; purchase-order-export	computed measure	4.7% of total PO value is maverick spend	VERIFIED

REPORT 03 OF 06

Transformation Recommendation

Procure-to-Pay · Acme Manufacturing

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Contents

1	Transformation intent
1.1	Guiding principles
2	Value vs. feasibility
3	Recommendations
4	Sequencing & readiness
4.1	Prioritization rationale
4.2	Implementation roadmap
4.3	Sequencing rationale
4.4	Strategic readiness
5	Success metrics
6	Risk register
7	Appendix — traceability matrix

Transformation Recommendation

Which opportunities to pursue, in what order, by value and feasibility.

1 Transformation intent

The procurement function operates under enforced, system-level controls that close the three governance gaps surfaced in the discovery data. First, the dual-approval rule for high-value purchases — under which any purchase order above EUR 50,000 requires the Category Manager plus a second Finance approval before release — is enforced automatically rather than by manual discretion. Today 7 purchase orders worth EUR 557,000 (12 percent of total PO value) were released on a single approval, every one of them above the EUR 50,000 threshold. In the target state, no PO above the threshold can be released without both approvals captured. Second, maverick spend is eliminated. The policy states that ordering without a purchase order is not permitted under any circumstances, yet 3 POs worth EUR 216,000 (4.7 percent of total PO value) were ordered before a PO existed, with 2 of those raised by a single buyer (buyer_a). In the target state, goods cannot be ordered ahead of an approved PO, removing this exposure at source and addressing the concentration in one buyer's activity. Third, three-way match evidence is restored to the procurement record. Policy requires POs to be matched against the goods receipt and supplier invoice, but the current PO export carries no goods-receipt or invoice column, leaving EUR 4,625,800 of PO value without match evidence. In the target state, every PO is traceable to receipt and invoice data so that the three-way match can be evidenced rather than assumed.

1.1 Guiding principles

P1

Strategic direction

Controls-first remediation: close the highest-value approval and maverick-spend gaps quickly, then restore three-way match evidence, then embed prevention into systems and governance.

P2

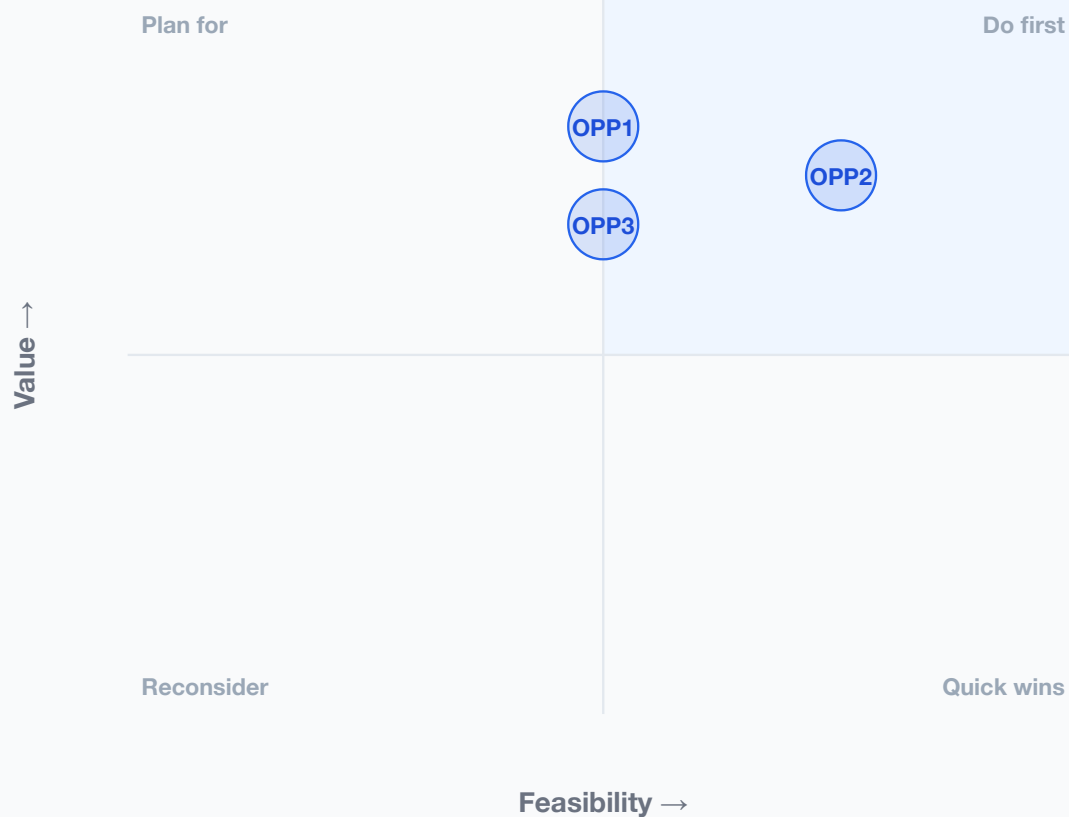
Sequencing

Sequencing is driven by exposure size and structural dependency.

P3

Readiness

Readiness is mixed.

2 Value vs. feasibility**WHERE EACH OPPORTUNITY SITS ON VALUE VERSUS FEASIBILITY****DO FIRST**

Address: Three-way match evidence cannot be verified across the PO population

Address: High-value purchase orders released on a single approval

PLAN FOR

Address: Maverick spend — goods ordered without a prior purchase order

CONSIDER*None in this quadrant***DEPRIORITISE***None in this quadrant***3 Recommendations**

Each opportunity as a recommendation — what it addresses, the phased actions, and what success looks like.

**R
01****Address: Three-way match evidence cannot be verified across the PO population****CRITICAL PRIORITY****HITL WORKFLOW**

Addresses PP1 | Delivers OPP1

The procurement policy requires a three-way match — purchase order, goods receipt, and supplier invoice — before an invoice is approved for payment. However, the available purchase order export does not carry the linked goods receipt and invoice evidence in a verifiable, structured form. As a result, the control that the policy mandates cannot be demonstrated across the purchase order population: reviewers cannot confirm, line by line, that each payment was supported by a matched receipt and invoice. This opportunity establishes a human-in-the-loop verification workflow that surfaces every purchase order alongside its matching evidence, routes exceptions to a reviewer, and produces an auditable record that the three-way match was performed.

Phased actions

- H1** The full purchase order population is loaded from the export into a single review workspace.
- H1** Each purchase order is automatically paired with its goods receipt and invoice records to form a candidate three-way match.

- H1** Purchase orders whose evidence is missing or does not reconcile are flagged and queued for a human reviewer.
- H1** A reviewer confirms matched orders and resolves or escalates exceptions, recording the decision.

Success criteria

Share of the purchase order population with a complete, verifiable three-way match

Number of exceptions routed to and cleared by reviewers

Time to resolve a flagged exception

Proportion of payments with an auditable match record at point of approval

**R
02**

Address: High-value purchase orders released on a single approval

CRITICAL PRIORITY

HITL WORKFLOW

Addresses PP3 | Delivers OPP2

High-value purchase orders are being released into the order pipeline on the strength of a single approval, bypassing the segregation-of-duties control that the procurement policy expects for spend above the established threshold. This opportunity introduces a human-in-the-loop dual-approval workflow that routes any purchase order at or above the high-value threshold to a second, independent approver before release. The aim is to close a control gap that currently allows material commitments to be authorised by one individual, restoring the layered sign-off the policy intends while keeping the approval path fast and auditable.

Phased actions

- H1** A requester creates a purchase order for goods or services in the procurement system.
- H1** The system checks the order value and flags any purchase order at or above the high-value threshold for dual approval.
- H1** The primary approver reviews and authorises the purchase order as today.
- H1** High-value orders are routed to a separate, independent approver who must sign off before release.

Success criteria

Share of high-value purchase orders released with two independent approvals reaches the target level

Number of high-value orders released on a single approval falls to zero

Complete dual-approval audit trail available for every high-value order

No increase in average approval cycle time beyond the agreed tolerance

R
03**Address: Maverick spend — goods ordered without a prior purchase order****CRITICAL PRIORITY****HITL WORKFLOW**

Addresses PP2 | Delivers OPP3

Maverick spend occurs when goods or services are ordered without a purchase order being raised and approved beforehand, meaning the commitment is created outside the controlled procurement workflow.

Phased actions

- H1** A staff member identifies a need and enters a requisition into the procurement workflow.
- H1** A purchase order is created and routed for approval before any order is placed with the supplier.
- H1** Incoming orders and invoices are screened to flag any commitment that has no prior approved purchase order.
- H1** A reviewer examines each flagged order, confirms whether it is genuine maverick spend, and decides on remediation.

Success criteria

Reduction in the count of invoices matched to retrospectively raised purchase orders

Proportion of orders with an approved purchase order dated before the order date

Number of flagged maverick-spend cases reviewed and remediated

Reduction in repeat maverick-spend offenders after coaching and escalation

4 Sequencing & readiness**4.1 Prioritization rationale**

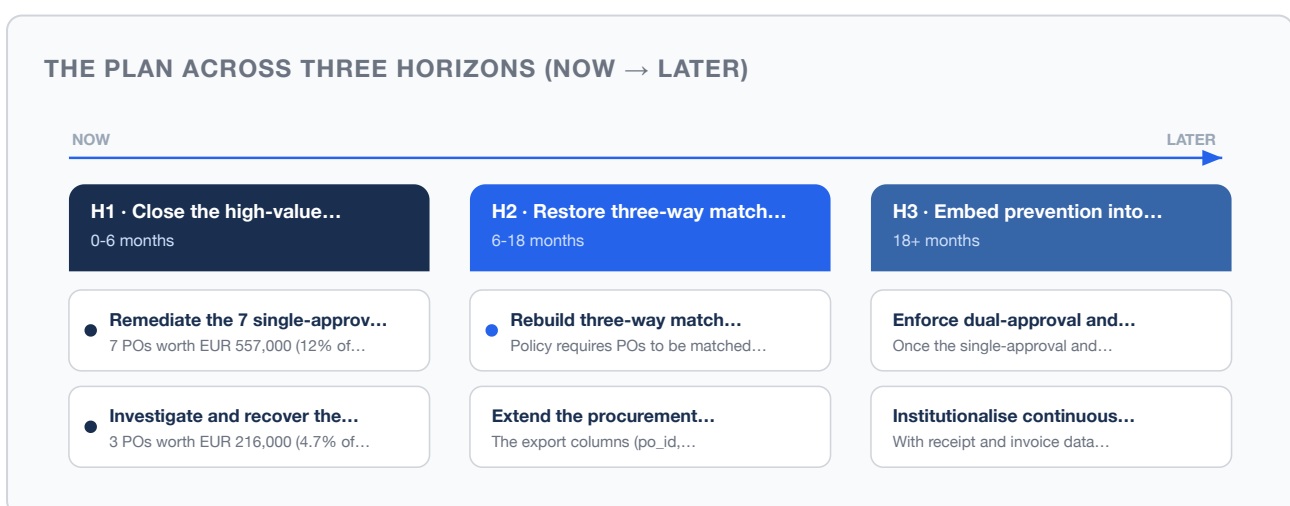
Each opportunity assessed across three readiness dimensions. The rating (high / medium / low) is followed by the reason, so the sequence is defensible rather than asserted.

OPPORTUNITY	DATA READINESS	TECHNICAL COMPLEXITY	OPERATIONAL READINESS
Address: Three-way match evidence cannot be verified across the PO population	THE PURCHASE ORDER EXPORT PROVIDES THE POPULATION, BUT GOODS RECEIPT AND INVOICE EVIDENCE IS NOT CURRENTLY JOINED TO IT IN A VERIFIABLE, STRUCTURED FORM; DATA LINKAGE WORK IS A PREREQUISITE.	MODERATE the core challenge is reliably linking purchase orders to goods receipts and invoices and persisting an audit trail, rather than complex automation logic.	A REVIEWER ROLE AND EXCEPTION-HANDLING PROCESS MUST BE ESTABLISHED BEFORE ROUTING CAN GO LIVE; THE POLICY ALREADY DEFINES THE UNDERLYING CONTROL.
Address: High-value purchase orders released on a single approval	THE PURCHASE-ORDER EXPORT IS AVAILABLE TO IDENTIFY THE AFFECTED HIGH-VALUE ORDER POPULATION, BUT THE THRESHOLD AND AFFECTED COUNTS MUST BE MEASURED BEFORE BENEFITS CAN BE QUANTIFIED.	LOW TO MODERATE value-based routing and dual-approval steps are standard configuration in most procurement workflow tools and do not require custom development.	REQUIRES CONFIRMED APPROVER ROLE ASSIGNMENTS AND A BACKUP APPROVER PATH BEFORE ACTIVATION; THE CONTROL REUSES THE EXISTING APPROVAL PROCESS AND ADDS A SECOND STEP.

OPPORTUNITY	DATA READINESS	TECHNICAL COMPLEXITY	OPERATIONAL READINESS
Address: Maverick spend — goods ordered without a prior purchase order	THE PURCHASE ORDER EXPORT IS AVAILABLE AND THE PROCUREMENT POLICY DEFINES THE REQUIRED APPROVE-THEN-ORDER SEQUENCE. RELIABILITY OF ORDER DATE VERSUS PURCHASE ORDER APPROVAL DATE IN THE EXPORT SHOULD BE CONFIRMED BEFORE DETECTION LOGIC IS FINALISED, AND MAVERICK-SPEND VOLUME AND VALUE MUST BE QUANTIFIED FROM THE EXPORT BEFORE NUMERIC TARGETS ARE COMMITTED.	MODERATE the core capability is matching orders and invoices against prior approved purchase orders in existing systems, with a review queue for exceptions.	A PROCUREMENT REVIEWER AND AN INVOICE-MATCHING CONTROL IN ACCOUNTS PAYABLE ARE NEEDED TO OPERATE THE HUMAN-IN-THE-LOOP STEP. ESCALATION AND COACHING ROUTES FOR REPEAT OFFENDERS SHOULD BE AGREED WITH PROCUREMENT LEADERSHIP.

4.2 Implementation roadmap

The recommendations sequenced across three horizons (now → later).



4.3 Sequencing rationale

Sequencing is driven by exposure size and structural dependency. The single-approval gap is the largest quantified control breach by value (EUR 557,000, 12 percent of total PO value).

value) and maps directly to an existing, unambiguous policy threshold, so enforcing dual approval above EUR 50,000 is the natural first move — it is a rule already written, simply not enforced. Maverick spend follows: it is smaller in value (EUR 216,000, 4.7 percent) but is a categorical policy breach ('not permitted under any circumstances') and is concentrated, with 2 of 3 cases from one buyer, making it tractable through targeted control and coaching. The three-way match data gap is sequenced as a structural enabler running in parallel, because the absence of goods-receipt and invoice columns (EUR 4,625,800 of value lacking match evidence) undermines the assurance value of the first two fixes — approval and pre-order controls are only fully verifiable once match evidence is captured in the record.

Dependencies: The maverick-spend and single-approval remediations depend on enforcement being moved from manual judgement into the procurement system's release workflow. The three-way match remediation depends on integrating or surfacing goods-receipt and invoice data alongside the PO record — without those two data elements the match cannot be evidenced regardless of approval discipline. Buyer-level remediation for maverick spend depends on the concentration insight (2 of 3 cases from buyer_a) being available to the accountable role for targeted follow-up.

4.4 Strategic readiness

Readiness is mixed. The organisation has a clear, written procurement policy covering all three control areas — dual approval above EUR 50,000, mandatory PO-before-order, and three-way match — so the control intent already exists and does not need to be designed from scratch. The discovery export demonstrates the data can be interrogated to surface breaches by count, value and buyer. The principal readiness gap is evidential and systemic: the PO export lacks goods-receipt and invoice columns, so match assurance cannot currently be produced from the procurement record. This is an amber condition affecting EUR 4,625,800 of PO value and should temper confidence in any claim that controls are operating until the data gap is closed.

5 Success metrics

How to measure delivery once live — the baseline today and the directional target.

METRIC	WHAT IT MEASURES	TARGET
Single-approval breach value	Total EUR value of purchase orders above the EUR 50,000 threshold released without the required second (Finance) approval. Baseline: EUR 557,000 across 7 POs, equal to 12 percent of total PO value.	Reduce toward zero — no above-threshold PO released on a single approval.
Maverick-spend value	Total EUR value of purchase orders where goods were ordered before a PO existed (po_before_order=no). Baseline: EUR 216,000 across 3 POs, equal to 4.7 percent of total PO value, with 2 POs from buyer_a.	Reduce toward zero — eliminate ordering ahead of an approved PO.
PO value lacking match evidence	Total EUR value of POs that cannot be evidenced against goods receipt and supplier invoice because the procurement record lacks the supporting columns. Baseline: EUR 4,625,800.	Reduce toward zero as goods-receipt and invoice data are made available for matching.
Buyer concentration of maverick spend	Share of maverick-spend POs attributable to a single buyer. Baseline: 2 of 3 maverick POs raised by buyer_a.	Reduce concentration through targeted control and coaching of the highest-exposure buyer.

6 Risk register

The delivery risks, how likely and how serious each is, and how it is mitigated.

RISK	LIKELIHOOD	IMPACT	MITIGATION	OWNER
Dual-approval rule above EUR 50,000 remains manually enforced, allowing further single-approval releases to recur as seen in the 7 POs worth EUR 557,000.	HIGH	HIGH	Move the EUR 50,000 dual-approval rule into the system release workflow so above-threshold POs cannot be released without both Category Manager and Finance approval captured.	Head of Procurement
Maverick spend continues at source, repeating the EUR 216,000 ordered-before-PO pattern, with concentration in individual buyers.	MEDIUM	HIGH	Block ordering ahead of an approved PO in the system and run targeted coaching for the highest-exposure buyer (buyer_a, 2 of 3 cases).	Procurement Operations Manager

RISK	LIKELIHOOD	IMPACT	MITIGATION	OWNER
Three-way match cannot be evidenced because the procurement record lacks goods-receipt and invoice columns, leaving EUR 4,625,800 of PO value unassured.	HIGH	HIGH	Integrate goods-receipt and invoice data into the PO record so the three-way match can be produced and audited from source.	Finance Controller
Control improvements are reported as complete before match evidence exists, creating false assurance over the EUR 4,625,800 amber population.	MEDIUM	MEDIUM	Withhold any 'controls operating' assertion until the data gap is closed and breach metrics are reproducible from the record.	Internal Audit Lead

7 Appendix — traceability matrix

Each pain point traced through to the recommendation, opportunity, expected outcome and horizon that addresses it.

PAIN POINT	SUMMARY	SEVERITY	RECOMMENDATION	OPPORTUNITY	EXPECTED OUTCOME	HORIZON
7 purchase orders worth EUR 557,000 (12 percent of total PO value) were released on a single approval, every one above the EUR 50,000 dual-approval threshold.	Largest quantified control breach by value; rule already written, not enforced.	High	Enforce the policy's dual-approval rule (Category Manager plus Finance) for all POs above EUR 50,000 inside the system release workflow.	Restore enforced segregation of approval authority on the highest-value purchases.	No above-threshold PO releasable without both required approvals; single-approval breach value driven toward zero.	Near-term

PAIN POINT	SUMMARY	SEVERITY	RECOMMENDATION	OPPORTUNITY	EXPECTED OUTCOME	HORIZON
3 POs worth EUR 216,000 (4.7 percent of total PO value) were ordered before a PO existed, with 2 raised by buyer_a, breaching the policy that maverick spend is not permitted under any circumstances.	Categorical policy breach, smaller value, concentrated and tractable.	High	Block ordering ahead of an approved PO in the system and apply targeted control and coaching to the highest-exposure buyer.	Eliminate off-process spend at source and address buyer-level concentration.	Maverick-spend value driven toward zero; reduced single-buyer concentration.	Near-term
EUR 4,625,800 of PO value lacks three-way match evidence because the PO export carries no goods-receipt or invoice column, despite policy requiring match against receipt and invoice.	Amber data gap that underpins assurance for the other two fixes.	High	Integrate goods-receipt and supplier-invoice data into the procurement record so the three-way match can be evidenced and audited.	Make control operation provable from source rather than assumed.	PO value lacking match evidence reduced toward zero; approval and pre-order controls become verifiable.	Parallel structural enabler

REPORT 04 OF 06

AI Opportunity Portfolio

Procure-to-Pay · Acme Manufacturing

REPORT REFERENCE	Report 04 of 06
CLIENT	Acme Manufacturing
DOMAIN	Procure-to-Pay
PREPARED BY	AuroPro · Autonomous Discovery Platform
CLASSIFICATION	Confidential

Contents

1	Portfolio at a glance
2	Opportunities in detail

AI Opportunity Portfolio

The recommended interventions in full — what the problem is, how the process changes, and what it delivers.

1 Portfolio at a glance

OPPORTUNITY	PATTERN	WHO IT SERVES	KNOWLEDGE SOURCES	EXPECTED BEHAVIOUR
Address: Three-way match evidence cannot be verified across the PO population	HITL Workflow	Accounts Payable reviewer, Procurement compliance lead, Internal audit	Procurement policy three-way match requirement, Purchase order export population	Every purchase order should be paired with its goods receipt and invoice, exceptions should be surfaced for h...
Address: High-value purchase orders released on a single approval	HITL Workflow	Requester, Approver, Second approver, Finance	Procurement policy approval thresholds, Purchase order export	Every purchase order at or above the high-value threshold is held until a second, independent approver signs ...
Address: Maverick spend — goods ordered without a prior purchase order	HITL Workflow	Requester, Procurement reviewer, Finance/Accounts Payable	procurement-policy, purchase-order-export	Every order with a commitment value is preceded by an approved purchase order; any exceptions are detected, r...

2 Opportunities in detail

Address: Three-way match evidence cannot be verified across the PO population

HITL WORKFLOW

The procurement policy requires a three-way match — purchase order, goods receipt, and supplier invoice — before an invoice is approved for payment. However, the available purchase order export does not carry the linked goods receipt and invoice evidence in a verifiable, structured form. As a result, the control that the policy mandates cannot be demonstrated across the purchase order population: reviewers cannot confirm, line by

line, that each payment was supported by a matched receipt and invoice. This opportunity establishes a human-in-the-loop verification workflow that surfaces every purchase order alongside its matching evidence, routes exceptions to a reviewer, and produces an auditable record that the three-way match was performed.

TODAY

1. Raise purchase order

Requester · Procurement system

A requester creates a purchase order which is recorded in the purchase order export.

PO is captured without a structured link to downstream receipt and invoice records

2. Receive goods and invoice

Receiving and Accounts Payable · Finance system

Goods are received and the supplier invoice arrives, but these records sit outside the verifiable export.

Goods receipt and invoice evidence is not joined back to the PO in a structured way

3. Attempt three-way match

Accounts Payable clerk · Manual / spreadsheet

A clerk tries to confirm the PO, receipt, and invoice agree before approving payment, as the policy requires.

Match evidence is incomplete or absent, so the control cannot be demonstrated

4. Approve and pay

Approver · Finance system

The invoice is approved for payment without a verifiable record that the three-way match was completed.

No auditable trail proving the policy control was satisfied

Risk of paying unmatched or duplicate invoices



WITH THE CHANGE

1. Ingest PO population

System · Verification workflow

The full purchase order population is loaded from the export into a single review workspace.

2. Assemble match evidence

System · Verification workflow

Each purchase order is automatically paired with its goods receipt and invoice records to form a candidate three-way match.

3. Route exceptions to reviewer

System · Verification workflow

Purchase orders whose evidence is missing or does not reconcile are flagged and queued for a human reviewer.

4. Human review and disposition

Accounts Payable reviewer · Verification workflow

A reviewer confirms matched orders and resolves or escalates exceptions, recording the decision.

5. Produce audit record

System · Verification workflow

An auditable record is generated showing that the three-way match was performed for each purchase order.

Business impact. Without verifiable three-way match evidence, the organisation cannot demonstrate compliance with its own procurement policy and is exposed to paying invoices that were never matched to a receipt and a purchase order. This creates audit findings, potential overpayment and duplicate-payment risk, and reliance on manual reconciliation that does not scale. Establishing a human-in-the-loop verification workflow restores a demonstrable control: every purchase order is paired with its evidence, exceptions are surfaced for review, and an auditable trail proves the match was completed before payment.

How we get there: Impact is described qualitatively because no verified numeric facts were supplied for this section; the procurement policy establishes the three-way match requirement and the purchase order export is the population against which evidence completeness must be measured.

How it's delivered. Stand up a verification workspace, link evidence to the PO population, route exceptions to reviewers, and persist an audit record.

Who uses it. Accounts Payable reviewer, Procurement compliance lead, Internal audit

Expected behaviour. Every purchase order should be paired with its goods receipt and invoice, exceptions should be surfaced for human review, and an auditable record should confirm the three-way match was performed before payment.

Escalation & human fallback. Unresolved or non-reconciling exceptions are escalated to the procurement compliance lead for disposition.

Sources. Procurement policy three-way match requirement, Purchase order export population

Formats. Purchase order export

Connects: Purchase order export source, Goods receipt records, Supplier invoice records

Success looks like:

- Share of the purchase order population with a complete, verifiable three-way match
- Number of exceptions routed to and cleared by reviewers
- Time to resolve a flagged exception
- Proportion of payments with an auditable match record at point of approval

Dependencies: Independent — can start immediately.

Risks:

- Goods receipt and invoice records may not be linkable to purchase orders without additional source data
- Historical purchase orders may lack evidence entirely, requiring a remediation backlog
- Reviewer capacity may be insufficient if the exception volume is high
- Match logic may produce false positives that mask genuine control gaps

Where this comes from: Procurement Policy and Purchase Order Export

Address: High-value purchase orders released on a single approval

HITL WORKFLOW

High-value purchase orders are being released into the order pipeline on the strength of a single approval, bypassing the segregation-of-duties control that the procurement policy expects for spend above the established threshold. This opportunity introduces a human-in-the-loop dual-approval workflow that routes any purchase order at or above the high-value threshold to a second, independent approver before release. The aim is to close a control gap that currently allows material commitments to be authorised by one individual, restoring the layered sign-off the policy intends while keeping the approval path fast and auditable.

TODAY

1. Requisition raised

Requester · Procurement system

A requester creates a purchase order for goods or services in the procurement system.

2. Single approver reviews

Approver · Procurement system

The purchase order is routed to one approver who reviews and authorises it.

No second reviewer for high-value spend

Segregation-of-duties control not enforced

3. Order released

Procurement system · Procurement system

On the single approval, the purchase order is released to the supplier without further checks.

High-value commitment released on one signature

Limited audit trail of independent review

4. Commitment recorded

Finance · Procurement system

The committed spend is booked against the budget once the order is live.

**WITH THE CHANGE****1. Requisition raised**

Requester · Procurement system

A requester creates a purchase order for goods or services in the procurement system.

2. Value-based routing

Procurement system · Procurement system

The system checks the order value and flags any purchase order at or above the high-value threshold for dual approval.

3. First approval

Approver · Procurement system

The primary approver reviews and authorises the purchase order as today.

4. Independent second approval

Second approver · Procurement system

High-value orders are routed to a separate, independent approver who must sign off before release.

5. Order released with audit trail

Procurement system · Procurement system

Once both approvals are recorded, the order is released and the dual sign-off is logged for audit.

6. Commitment recorded

Finance · Procurement system

The committed spend is booked against the budget with a complete approval history attached.

Business impact. Releasing high-value purchase orders on a single approval leaves material spend commitments without the independent second review the procurement policy is designed to enforce. This creates exposure to error, unauthorised commitment and fraud on the orders that carry the largest financial consequence, and it weakens the audit trail finance relies on. Introducing a value-based dual-approval step restores segregation of duties on exactly the orders where it matters most, improves auditability, and reduces the risk of an incorrect or unauthorised high-value commitment reaching a supplier. Because no verified quantitative values were supplied for this opportunity, the impact is presented qualitatively; the count and euro value of affected purchase orders should be measured from the purchase-order export before benefits are sized.

How we get there: No verified numeric facts were provided in the inputs for this opportunity, so no measured figures are stated. All quantification is deferred to a measurement step against the purchase-order export and the policy threshold.

How it's delivered. Configure a value-based routing rule in the procurement system so that any purchase order at or above the high-value threshold defined in the procurement policy is automatically held for a second, independent approval before release. Map approver roles so the second approver cannot be the same person as the first, enforcing segregation of duties in the workflow itself. Add the dual sign-off to the order's audit log so finance can evidence the control. Validate the rule against historical orders from the purchase-order export to confirm the routing fires on the correct population before go-live.

Who uses it. Requester, Approver, Second approver, Finance

Expected behaviour. Every purchase order at or above the high-value threshold is held until a second, independent approver signs off, with both approvals recorded in the audit trail before release.

Escalation & human fallback. Where the second approval is not provided within the agreed turnaround, the order is escalated to a designated backup approver rather than released.

Sources. Procurement policy approval thresholds, Purchase order export **Formats.** Procurement policy document, Purchase order export

Connects: Procurement system approval workflow, Audit logging

Success looks like:

- Share of high-value purchase orders released with two independent approvals reaches the target level
- Number of high-value orders released on a single approval falls to zero
- Complete dual-approval audit trail available for every high-value order
- No increase in average approval cycle time beyond the agreed tolerance

Dependencies: Independent — can start immediately.

Risks:

- Approval cycle time may increase if the second approver is unavailable, requiring a defined backup approver
- If the high-value threshold is set incorrectly, the control may either miss material orders or create unnecessary friction on routine ones
- Approver role mappings must be kept current to prevent the same person fulfilling both approvals
- Benefit sizing cannot be confirmed until the affected order population is measured from the export

Where this comes from: Procurement Policy and Purchase Order Export

Address: Maverick spend — goods ordered without a prior purchase order

HITL WORKFLOW

Maverick spend occurs when goods or services are ordered without a purchase order being raised and approved beforehand, meaning the commitment is created outside the controlled procurement workflow.

TODAY

1. Need identified by requester

Requester

A staff member identifies a need for goods or services and proceeds to source them directly.

2. Order placed with supplier

Requester

The requester contacts the supplier and places the order without first raising a purchase order.

No purchase order exists, so the commitment is unapproved and unbudgeted at the point of order

3. Goods or invoice received

Receiving/Finance

Goods arrive or an invoice is presented before any procurement control has been applied.

Finance must match an invoice to a purchase order that was never created

4. Retrospective purchase order raised

Procurement

A purchase order is created after the fact to allow the invoice to be paid, defeating the preventive control.

After-the-fact approval cannot prevent off-contract pricing or unauthorised commitments



WITH THE CHANGE

1. Need identified by requester

Requester · Procurement workflow

A staff member identifies a need and enters a requisition into the procurement workflow.

2. Purchase order raised and approved

Procurement · Procurement workflow

A purchase order is created and routed for approval before any order is placed with the supplier.

3. Maverick-spend detection

Detection control · Procurement workflow

Incoming orders and invoices are screened to flag any commitment that has no prior approved purchase order.

4. Human review of flagged orders

Procurement reviewer · Procurement workflow

A reviewer examines each flagged order, confirms whether it is genuine maverick spend, and decides on remediation.

5. Remediation and feedback

Procurement · Procurement workflow

Confirmed maverick spend is logged, the requester is coached, and recurring offenders are escalated to bring behaviour back into compliance.

Business impact. Maverick spend bypasses the preventive controls in the procurement policy: commitments are made before approval, off-contract pricing cannot be

challenged, and Finance is forced to raise retrospective purchase orders simply to pay invoices. Introducing a detection-and-review control surfaces these cases at the point of order rather than at invoice, restores the policy-mandated sequence of approve-then-order, and provides an audit trail of non-compliant purchasing. Because no verified figures for the volume or value of maverick spend were supplied to this section, the impact is described qualitatively and the underlying counts should be quantified from the purchase order export before targets are set.

How we get there: No numeric VERIFIED FACTS were provided for this section, so no measured counts, values, or percentages are stated. Quantification of maverick-spend volume and value is to be derived directly from the purchase-order-export once those figures are confirmed.

How it's delivered. Implement a screening control over the purchase order export and incoming invoices that identifies orders with no matching prior approved purchase order. Flagged items are routed to a procurement reviewer who confirms whether the order is genuine maverick spend, records the outcome, and triggers remediation. Pair the control with feedback to requesters and an escalation path for repeat offenders so the policy-mandated approve-then-order sequence becomes the default behaviour.

Who uses it. Requester, Procurement reviewer, Finance/Accounts Payable

Expected behaviour. Every order with a commitment value is preceded by an approved purchase order; any exceptions are detected, reviewed by a human, and remediated rather than silently paid through a retrospective purchase order.

Escalation & human fallback. Confirmed maverick-spend cases and repeat offenders are escalated through procurement leadership for remediation and policy enforcement.

Sources. procurement-policy, purchase-order-export **Formats.** Purchase order export

Connects: Procurement workflow / purchase order system, Accounts payable invoice matching

Success looks like:

- Reduction in the count of invoices matched to retrospectively raised purchase orders
- Proportion of orders with an approved purchase order dated before the order date
- Number of flagged maverick-spend cases reviewed and remediated
- Reduction in repeat maverick-spend offenders after coaching and escalation

Dependencies: Independent — can start immediately.

Risks:

- The purchase order export may not reliably capture order date versus purchase order approval date, limiting detection accuracy

- Reviewers may approve flagged items without genuine challenge, weakening the control
- Retrospective purchase orders could continue if Finance lacks an enforced block on unmatched invoices
- Volume of flagged items may exceed review capacity if the detection threshold is too broad

Where this comes from: Procurement Policy and Purchase Order Export

REPORT 05 OF 06

Transformation Roadmap

Procure-to-Pay · Acme Manufacturing

REPORT REFERENCE	Report 05 of 06
CLIENT	Acme Manufacturing
DOMAIN	Procure-to-Pay
PREPARED BY	AuroPro · Autonomous Discovery Platform
CLASSIFICATION	Confidential

Contents

1	Implementation roadmap
2	Horizon detail
3	Planning assumptions

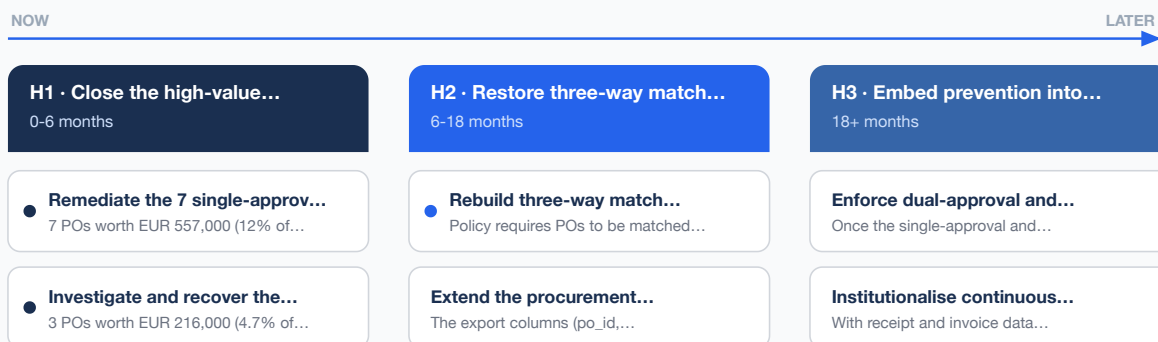
Transformation Roadmap

The recommended opportunities, sequenced across three horizons.

Strategic direction. Controls-first remediation: close the highest-value approval and maverick-spend gaps quickly, then restore three-way match evidence, then embed prevention into systems and governance.

1 Implementation roadmap

THE PLAN ACROSS THREE HORIZONS (NOW → LATER)



2 Horizon detail

H1 — Close the high-value approval and maverick-spend control breaches (0-6 months)

- Remediate the 7 single-approval purchase orders above the EUR 50,000 threshold** opportunity — 7 POs worth EUR 557,000 (12% of total PO value) were released on single approval, yet policy requires Category Manager AND Finance approval above EUR 50,000. All 7 values fall in {54000,61000,67000,72000,88000,95000,120000}, confirming every one breached the threshold. Retroactive secondary review closes the most concentrated, clearly-defined control gap first.
- Investigate and recover the 3 maverick-spend purchase orders** opportunity — 3 POs worth EUR 216,000 (4.7% of total PO value) were ordered with no prior PO, which policy prohibits under any circumstances. The spend concentrates on buyer_a (2 POs) and buyer_d (1), making remediation targeted and actionable.

H2 — Restore three-way match evidence and harden the data foundation (6-18 months)

- **Rebuild three-way match evidence for the EUR 4,625,800 exposure** opportunity
 - Policy requires POs to be matched against goods receipt and supplier invoice, but the export carries no goods_receipt or invoice column, leaving EUR 4,625,800 of PO value without match evidence (amber). This is the single largest exposure and depends on extending the data set before it can be assessed.
- **Extend the procurement export to capture receipt and invoice fields** — The export columns (po_id, supplier, category, amount_eur, raised_by, approval_status, po_before_order) omit goods_receipt and invoice, which is why the match-evidence figure is amber rather than verified. Capturing these fields is a prerequisite to closing the match-evidence gap with confidence.

H3 — Embed prevention into systems and governance (18+ months)

- **Enforce dual-approval and PO-before-order rules as system controls** — Once the single-approval and maverick-spend breaches are remediated, the EUR 50,000 dual-approval rule and the mandatory PO-before-order rule should be enforced automatically so that the patterns seen across 7 single-approval and 3 maverick POs cannot recur.
- **Institutionalise continuous three-way match monitoring** — With receipt and invoice data captured, three-way match should move from one-off remediation to ongoing monitoring, sustaining assurance over the value previously exposed at EUR 4,625,800.

3 Planning assumptions

The forward-looking elements of this roadmap — dates, owners, service levels, targets, cadence and sequence — are planning assumptions, not measured findings. They are shown here explicitly so they can be confirmed before delivery.

PLANNING ASSUMPTION — CONFIRM BEFORE DELIVERY

These forward-looking items are recommendations, not measured facts.

TYPE	ASSUMPTION	BASIS
SEQUENCING	Re-validate and obtain retrospective dual sign-off on the 7 single-approval POs above EUR 50,000 as the first remediation step.	7 POs worth EUR 557,000 flagged <i>single_approval_only</i> , all above the EUR 50,000 dual-approval threshold
OWNERSHIP	Hold a control review with buyer_a, who raised 2 of the 3 maverick-spend POs, to address off-process ordering.	Maverick POs raised by buyer_a = 2; <i>policy prohibits ordering without a PO</i>
SEQUENCING	Add goods-receipt and invoice fields to the procurement export so three-way matching can be evidenced.	EUR 4,625,800 of PO value lacks match evidence; export has no <i>goods_receipt</i> or <i>invoice</i> column
SEQUENCING	Add goods_receipt and invoice reference fields to the PO system export so three-way match can be evidenced and reported, closing the assurance gap on the EUR 4,625,800 currently unverifiable.	EUR 4,625,800 of PO value has no match evidence because no <i>goods_receipt</i> or <i>invoice</i> column is present in the export.
TARGET	Enforce a system-level block that prevents release of any PO above EUR 50,000 without both Category Manager and Finance approval.	Policy requires dual approval above EUR 50,000, yet 7 POs (all above the threshold) were released on a single approval.
OWNERSHIP	Prioritise buyer-level remediation starting with buyer_a, who raised 2 of the 3 maverick-spend orders.	buyer_a accounts for 2 of the 3 <i>po_before_order=no</i> cases.
SEQUENCING	Introduce a preventive control requiring a PO to exist before any supplier order is placed, removing the ability to create POs retrospectively.	3 POs were recorded with <i>po_before_order=no</i> , totalling EUR 216,000 of maverick spend.
SEQUENCING	Sequence the dual-approval enforcement first, maverick-spend blocking second, and the goods-receipt/invoice data integration in parallel as a structural enabler.	Single-approval breach is the largest quantified exposure (EUR 557,000, 12 percent); maverick spend is smaller (EUR 216,000, 4.7 percent); match data gap (EUR 4,625,800) underpins assurance for both.
OWNERSHIP	Assign the dual-approval workflow enforcement to the Head of Procurement.	7 single-approval POs above the EUR 50,000 policy threshold.

TYPE	ASSUMPTION	BASIS
OWNERSHIP	Assign maverick-spend blocking and buyer coaching to the Procurement Operations Manager.	3 maverick POs, 2 of them from buyer_a.
OWNERSHIP	Assign goods-receipt and invoice data integration to the Finance Controller.	EUR 4,625,800 PO value lacks match evidence; PO export has no goods-receipt or invoice column.
TARGET	Set a target of zero above-threshold POs released on a single approval, with periodic monitoring against the baseline.	All 7 single-approval POs exceed the EUR 50,000 policy threshold.
CADENCE	Establish a regular review cadence of breach metrics (single-approval, maverick, match-evidence) once data is reproducible from the record.	Breach counts and values are currently derivable from the PO export.
SEQUENCING	Defer any assertion that controls are operating until the match-evidence data gap is closed.	EUR 4,625,800 of PO value is flagged amber for lacking match evidence.
TIMING	H1 spans the first 0-6 months to prioritise the highest-value, policy-explicit breaches.	7 single-approval POs worth EUR 557,000 and 3 maverick POs worth EUR 216,000 are verified breaches
TIMING	H2 spans 6-18 months to allow data extension before assessing the match-evidence exposure.	EUR 4,625,800 of PO value lacks match evidence (amber) and the export has no goods_receipt or invoice column
TIMING	H3 covers 18+ months for systemic prevention and continuous monitoring.	policy mandates dual approval above EUR 50,000, PO-before-order, and three-way match
SEQUENCING	Retroactive secondary approval should be obtained for each single-approval PO before any further release to affected suppliers.	policy requires Category Manager AND Finance approval above EUR 50,000
OWNERSHIP	A named control owner (e.g. Head of Procurement) should sign off remediation of the maverick and single-approval POs.	maverick spend concentrated on buyer_a (2) and buyer_d (1)
TARGET	Set a zero-tolerance target threshold for new maverick spend once system enforcement is live.	policy states maverick spend is not permitted under any circumstances

TYPE	ASSUMPTION	BASIS
CADENCE	Three-way match monitoring should run on a recurring (e.g. monthly) cadence once receipt and invoice data is captured.	<i>policy requires PO-to-receipt-to-invoice three-way match</i>
COST	Extending the export schema and enforcing system controls will require IT/procurement-system investment to be budgeted.	<i>export columns currently omit goods_receipt and invoice fields</i>
OWNERSHIP	A named Accounts Payable owner should be accountable for the exception review queue.	<i>Three-way match approval responsibility defined in the procurement policy</i>
SERVICE LEVEL	Set a target service level for clearing flagged exceptions before payment approval.	<i>Policy requirement that the three-way match precede payment</i>
TARGET	Define a minimum acceptable threshold for the share of the PO population carrying verifiable match evidence.	<i>PO population available in the purchase order export</i>
CADENCE	Run evidence-completeness reporting on a regular cadence to monitor control health.	<i>Ongoing nature of the three-way match control in the procurement policy</i>
SEQUENCING	Begin with a read-only completeness assessment before enabling reviewer routing and audit records.	<i>Need to confirm data linkage reliability against the purchase order export</i>
TARGET	Define the high-value spend threshold that triggers mandatory dual approval, aligned to the procurement policy.	<i>procurement-policy specifies approval expectations for higher-value spend</i>
OWNERSHIP	Assign an accountable owner for maintaining approver role mappings and the routing rule.	<i>after_process requires an independent second approver distinct from the first</i>
SERVICE LEVEL	Agree a maximum turnaround time for the second approval and a backup approver where the SLA is at risk.	<i>risk that the second approval step lengthens cycle time</i>

TYPE	ASSUMPTION	BASIS
SEQUENCING	Validate the routing rule against historical export data before enabling it in production.	<i>purchase-order-export provides historical orders to test routing</i>
TARGET	Define a minimum commitment value above which orders without a prior purchase order are flagged for human review.	<i>Procurement policy requires a purchase order before goods are ordered.</i>
OWNERSHIP	Assign a named procurement reviewer to triage and remediate flagged maverick-spend cases.	<i>After-process human review step requires an accountable reviewer.</i>
SERVICE LEVEL	Set a target turnaround for reviewing each flagged order before invoice payment.	<i>Detection occurs at point of order/invoice in the after-process.</i>
CADENCE	Run the maverick-spend detection screen on a regular cycle against the purchase order export.	<i>Detection control operates over the purchase-order-export data.</i>
TARGET	Set a target reduction percentage for retrospective purchase orders against a measured baseline once volumes are quantified from the export.	<i>Retrospective purchase orders are evidenced in the before-process.</i>

REPORT 06 OF 06

Supporting Artefacts

Procure-to-Pay · Acme Manufacturing

REPORT REFERENCE	Report 06 of 06
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Contents

1	Source provenance
2	Success metrics framework
3	System & data-flow map

Supporting Artefacts

Reference material behind this assessment.

1 Source provenance

Every figure in this assessment is computed from the sources below and traces back to them.

SOURCE	TYPE	WHAT IT IS	FINDINGS IT SUPPORTED
Purchase Order Export	System export	your Purchase Order Export	F1, F2, F3
Procurement Policy	Process documentation	your Procurement Policy	F1, F2, F3

2 Success metrics framework

How the impact of the recommended interventions should be measured once live — the dimension, what it means, and the target to hold delivery to.

METRIC	DEFINITION	TARGET
Single-approval breach value	Total EUR value of purchase orders above the EUR 50,000 threshold released without the required second (Finance) approval. Baseline: EUR 557,000 across 7 POs, equal to 12 percent of total PO value.	Reduce toward zero — no above-threshold PO released on a single approval.
Maverick-spend value	Total EUR value of purchase orders where goods were ordered before a PO existed (po_before_order=no). Baseline: EUR 216,000 across 3 POs, equal to 4.7 percent of total PO value, with 2 POs from buyer_a.	Reduce toward zero — eliminate ordering ahead of an approved PO.
PO value lacking match evidence	Total EUR value of POs that cannot be evidenced against goods receipt and supplier invoice because the procurement record lacks the supporting columns. Baseline: EUR 4,625,800.	Reduce toward zero as goods-receipt and invoice data are made available for matching.

METRIC	DEFINITION	TARGET
Buyer concentration of maverick spend	Share of maverick-spend POs attributable to a single buyer. Baseline: 2 of 3 maverick POs raised by buyer_a.	Reduce concentration through targeted control and coaching of the highest-exposure buyer.

3 System & data-flow map

The systems view of the same domain — which business systems the process touches and how data moves between them.

A full technical trace of every figure in this assessment is available to your data team on request.